

Vinay Chintala

Process engineer

Tallahassee, Florida | vinaychemy@gmail.com | +1 4482007987 | vinaychemy@gmail.com | LinkedIn | Git hub

SUMMARY

Chemical Engineering M.S. student at Florida State University with experience in wastewater treatment, petroleum operations, pharmaceutical manufacturing, and compliance style documentation. Skilled in environmental compliance, quality assurance, performance tracking, technical reporting, data analysis, process safety, and MS Office. Strong background in water treatment, production records, process improvement, and petroleum process systems.

EXPERIENCE

Research Assistant

Nov 2025 – May 2026

Florida State University | Tallahassee, Florida

- Tested polymer samples in a materials research lab to study how they behave under heat and mechanical stress.
- Operated lab equipment including 3D printers and tested machines to support ongoing research experiments.
- Prepared samples and maintained equipment calibration to ensure consistent and accurate test results.

Process Engineering Apprentice

Oct 2023 – Mar 2024

Dr. Reddy's Laboratories | Telangana, India

- Maintained batch records covering Good Manufacturing Practice, safety, data integrity, and production control.
- Reviewed capacity-improvement work that increased monthly output from 135 to 165 tonnes.
- Tracked cycle-time changes that reduced key batch steps from 16 to 12 hours.
- Assessed wash-water reduction that lowered usage from 15,000 to 10,000 liters per batch. pharmaceutical batch manufacturing from intermediate-stage processing to finished product handling.

Research Assistant

Jan 2023 – Sep 2023

Osmania University | Hyderabad, India

- Developed a MATLAB-based control model to regulate liquid levels in a dual-tank system.
- Tuned the model to improve control accuracy by 58% and reduce settling time by 15%.
- Enhanced system stability over a baseline PID controller with 23% faster response and 98% lower overshoot.

Process Engineering Intern

May 2022 – Oct 2022

Oil and Natural Gas Corporation Ltd | Odalarevu, India

- Reviewed 7 million cubic meters/day onshore gas terminal operations supporting petroleum and gas processing.
- Examined 3-stage compressor systems used for pressure control, gas movement, and process reliability.
- Studied glycol drying and regeneration practices used to control moisture, corrosion, and product quality.
- Documented chemical recovery, water treatment, safety compliance, alarms, and emergency response procedures.

Environmental Engineering Intern

Mar 2022 – May 2022

Terra Green Technologies Pvt. Ltd | India

- Covered industrial and urban wastewater treatment, including sewage, effluent, and reverse-osmosis systems.
- Compared physical, biological, and advanced methods used for industrial effluent treatment.
- Interpreted a case study involving oil-water separation, pH control, coagulation, and solids removal.
- Estimated pump requirements for water-treatment service, including suction head and shaft power.

PROJECTS

Heavy Oil Recovery and Refinery Integration Study

Feb 2026 – May 2026

- Reviewed heavy crude oil recovery methods, including steam injection and sand-assisted production techniques.
- Compared pipeline and partial-upgrading options for high-viscosity crude with transport and refinery limits.
- Built a 3-stage scoring framework to evaluate recovery routes across shallow, deep, and remote oil-field cases.

Experimental and Numerical Analysis of Food Dye Diffusion in Water and Gelatin

Sep 2025 – Dec 2025

- Measured food-dye diffusion in water and gelatin using ImageJ and MATLAB.
- Led a 5 member team on radial diffusion modeling, reaching 89% accuracy for gelatin and 77% for water.

Sodium Lauryl Sulfate Plant Design

Jun 2022 – Apr 2023

- Designed a 150 kg/day Sodium Lauryl Sulfate plant involving sulphonation, neutralization, and spray drying.
- Performed material and energy balances to size reactors, storage tanks, and centrifugal pumps.
- Specified jacketed reactor design using heat-transfer area, shell thickness, and agitator requirements.
- Projected \$4.56 million capital investment and 2.36 year payback with site selection in Visakhapatnam.

EDUCATION

Florida State University <i>M.S. in Chemical Engineering GPA: 3.5/4.0</i>	<i>Jan 2025 – Dec 2026</i>
Osmania University <i>B.Tech in Chemical Engineering GPA: 3.2/4.0</i>	<i>Jul 2019 – May 2023</i>

TECHNICAL SKILLS

Environmental & Compliance: Environmental Compliance, Wastewater Treatment, Quality Assurance, Technical Documentation, Process Safety, Data Integrity
Petroleum & Process Systems: Petroleum Operations, Natural Gas Operations, Water Treatment, Chemical Recovery, Production Control, Process Optimization
Data & Reporting: Performance Metrics, Data Analysis, Root Cause Analysis, Case Analysis, Numerical Modeling, Six Sigma
Tools & Software: MS Office, Excel, Word, PowerPoint, Aspen Plus, Aspen HYSYS, MATLAB, AutoCAD P&ID, Python, Simulink

CERTIFICATIONS

SACHe Managing Lab Process Safety 1 & 2	<i>May 2026</i>
Six Sigma: Green Belt	<i>Jan 2026</i>
MATLAB Onramp	<i>Aug 2025</i>
Computational Process Design	<i>Apr 2024</i>
Aspen HYSYS	<i>Nov 2023</i>
Aspen Plus Simulation	<i>Oct 2023</i>
Nanomaterials for Energy Harvesting and Storage Applications	<i>Aug 2023</i>
Simulink Onramp	<i>Jun 2023</i>

ACHIEVEMENTS & AWARDS

GATE Exam 2023 Qualified – Secured All India Rank 2021 in Chemical Engineering.
Medha Charitable Trust Merit Scholar – Selected through a state-level merit exam in India.